Why is the Bayesian network so confident?

Diagnosis and plan for a defensible Bayesian uncertainty for `HEFTNet`

For approval before any code is written

October 31, 2026

Abstract

The full-data Bayesian network (`runs/BNN_poisson_full`) gives a posterior mean almost as accurate as the fine-tuned ensemble, but a posterior standard deviation of only 0.5% of the POWHEG noise, so that its pulls have a standard deviation of 1.37 just above threshold. Before the Bayesian network appears in a paper, we must know *why*. A posterior that is too narrow can have four causes: (A) the chain has not explored the mode it sits in, (B) it has not visited the other modes, (C) the likelihood claims the network describes POWHEG better than it does, and (D) the prior. A and B are failures of the computation; C and D are choices in the model. Only once all four are understood can we say what Bayes' theorem actually delivers here. Evidence already in hand points strongly to A, and the model-error study of the appendix shows that C is real. This plan proposes five steps, cheapest first, each with a decision rule: a linearised Laplace approximation that gives the within-mode width *without sampling*; HMC with a dense mass matrix taken from it; several chains from different starting points; a likelihood with a sampled model-error term; and a prior-sensitivity check. Nothing is written until you approve; the decisions are collected in section 12.

The message for the paper

Bayes' theorem is not on trial. The posterior answers a precisely posed question: given *this* network, *this* prior and a likelihood that says the POWHEG noise is the only discrepancy, how well are the network's parameters, and hence its predictions, determined? If the computation is done correctly, that answer is right, and if the predictions are then too confident, the fault lies in the question, namely in the likelihood, which ignores the error of the model itself. The plan is designed to end in one of two defensible statements: either "with an adequate sampler and a likelihood that includes a model-error term, the Bayesian predictive uncertainty is calibrated and agrees with the ensemble", or a clear account of which of the four causes prevents this.

Contents

1 What we observed **2**

2 Four ways a posterior can be too narrow **2**

3 Evidence already in hand **3**

3.1 An identity: the effective number of parameters 3

3.2 The chain was still relaxing 4

3.3 The step size . . . 4
3.4 The likelihood is too confident . . . 5

4 Step 1: the linearised Laplace approximation **5**
4.1 What it is . . . 5
4.2 What it delivers . . . 5

5 Step 2: HMC with a dense mass matrix **6**

6 Step 3: several chains **7**

7 Step 4: a likelihood that admits the model error **7**

8 Step 5: sensitivity to the prior **8**

9 What could go in the paper **8**

10 Code **9**

11 Order of work **9**

12 Decisions for you **9**

1 What we observed

	median s/σ_k	s.d. of pull, 255–555 GeV	s.d. of pull, $>$ 555 GeV
Bayesian network (`BNN_poisson_full`)	0.005	1.37	1.06
fine-tuned ensemble, bootstrap	0.056	1.15	1.04
+ model error, $a(m_{hh})$	0.32	1.02	1.00

Table 1: Calibration on the 1,000 test spectra (from the draft appendix).

The run: Gaussian likelihood with the known POWHEG variances (the `poisson` option), hierarchical Gaussian prior with 16 Gibbs-sampled precisions [2], warm start from `runs/L1Loss`, 1,000 warm-up iterations, 1,000 samples, 800 leapfrog steps, acceptance rate 0.72, final step size $\varepsilon \approx 7.7 \times 10^{-4}$. The effective sample size of the two monitored predictions is about 135. The posterior mean differs from the ensemble mean by a median of 0.8 ensemble standard deviations: a perfectly sensible central value, with an implausibly narrow band around it.

2 Four ways a posterior can be too narrow

A. Poor sampling within a mode. HMC moves along stiff and soft directions alike with one step size, set by the stiffest. If the mass matrix does not equalise the curvatures, the soft directions are explored by a slow random walk, and a chain of 10^3 iterations sees only a fraction of their width.

B. Modes not visited. Networks have many separated modes (equally good solutions that are not related by a symmetry). A single chain stays in one [8]. The seeds ensemble shows that about 80% of the ensemble variance is of this kind.

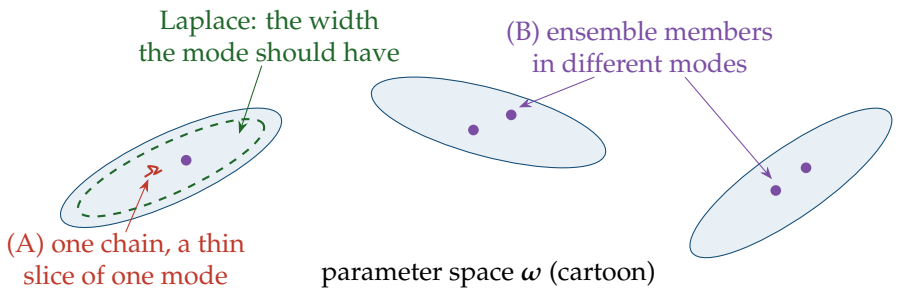


Figure 1: Cartoon of the posterior: several separated modes (blue). A chain that mixes poorly covers a thin slice of one mode (red); the linearised Laplace approximation gives the width a well-sampled mode should have (green); ensemble members sit in different modes (purple). Causes C and D change the shape of the ellipses themselves.

C. An over-confident likelihood. The likelihood $\prod_k \mathcal{N}(t_k; f_k, \sigma_k^2)$ asserts that the network can represent the true cross section exactly and that the POWHEG noise is the only discrepancy. The model-error study says otherwise: on data not used in the training, a further variance $a\,\sigma_k^2$ with $a = 0.1$–0.75 is needed.

D. The prior. With 2.7×10^5 training rows the likelihood dominates every direction that affects the predictions at points like the training points; the prior mainly controls the directions the data leave free, which hardly affect the predictions. A factor of ten from the prior would be surprising, but it is cheap to check.

A and B are different questions

Even a perfect sampler need not make the posterior as wide as the ensemble. Whether the modes of fig. 1 contribute to the posterior depends on their probability *mass*: if one mode has a mass e^{100} times that of the others, the posterior is that mode, and the ensemble spread, although a fair description of the variability of the *training procedure*, is not part of the Bayesian answer. Several chains tell us which case we are in (section 6).

3 Evidence already in hand

3.1 An identity: the effective number of parameters

For a model linear in its parameters, $f = J\omega$, with Gaussian noise of known variances σ_k^2 and a Gaussian prior of precision A, the posterior covariance is $\Sigma = (J^\mathsf{T} W J + A)^{-1}$ with $W = \mathrm{diag}(1/\sigma_k^2)$, and

$$\sum_{k=1}^{N} \frac{\mathrm{Var}_{\mathrm{post}}[f_k]}{\sigma_k^2} = \mathrm{tr}\big(J^\mathsf{T} W J \Sigma\big) = \sum_i \frac{\lambda_i}{\lambda_i + \alpha_i} \equiv \gamma, \tag{1}$$

where, for a prior $A = \alpha I$, the λ_i are the eigenvalues of $J^\mathsf{T} W J$. This is MacKay's "number of well-determined parameters" [1]: each direction pinned down by the data contributes one, each direction left to the prior contributes nothing, so $0 \le \gamma \le p$. Near a mode a network is approximately linear in its parameters, so eq. (1) applies approximately: the sum over the training rows of s_k^2/σ_k^2, with s_k the posterior standard deviation of the prediction, counts the well-determined parameters.

What the run says

Estimating the sum as N times the average of s_k^2/σ_k^2 over the test spectra (right panel of fig. 2), the Bayesian run gives $\gamma \approx 50$ out of $p = 3{,}944$ parameters. For a network that represents 23 functions of m_{hh} well enough to give $\chi^2/N \approx 1.1$ on 80 bins over a 5-dimensional parameter space, that is far too few. The fine-tuned ensembles, which are not posteriors, give 3,100–4,800: the order of magnitude a well-determined network of this size should have. A correctly sampled single mode is therefore expected to give a width of order the ensemble spread, roughly ten times what the chain reports. This points to cause A.

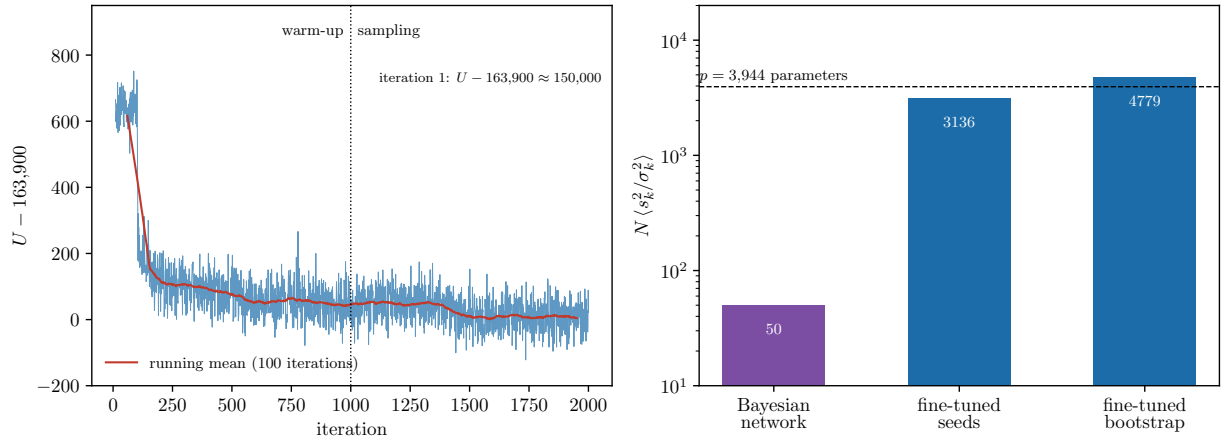


Figure 2: Left: the potential energy U of `BNN_poisson_full` from iteration 10 on (it starts 150,000 higher, at the L1 solution, and falls within ten iterations). It falls in steps, at the first mass-matrix updates, and is still drifting downwards during sampling, by about 100 over 1,800 iterations, two standard deviations of its fluctuations. Right: $N\langle s_k^2/\sigma_k^2\rangle$, an estimate of the number of well-determined parameters, eq. (1), for the Bayesian network and, for comparison, for the two fine-tuned ensembles.

3.2 The chain was still relaxing

The left panel of fig. 2 shows that the chain reached the χ^2 region within ten iterations, sat nearly still for the first 150 iterations (the step size was $\sim 10^{-7}$ until the first mass-matrix updates), and has been drifting slowly downhill ever since. A drift that persists through the whole sampling phase means the chain has not reached its stationary distribution; samples drawn while it relaxes are too tightly clustered. Note that the size of the fluctuations of U (standard deviation 45, close to $\sqrt{p/2} = 44$ expected for a p-dimensional Gaussian) is *not* evidence of good mixing: they are dominated by the stiff directions, which mix fast, and by the Gibbs updates of the 16 precisions.

3.3 The step size

The leapfrog integrator is stable only if $\varepsilon \lesssim 2/\sqrt{\lambda_{\max}}$, where $\lambda_{\max}$ is the largest curvature of U in the coordinates set by the mass matrix. If the diagonal mass matrix had equalised the curvatures, ε would be of order $p^{-1/4} \approx 0.1$ and a few tens of leapfrog steps would cross a mode. The tuned $\varepsilon \approx 8 \times 10^{-4}$ implies a remaining curvature ratio of order 10^6: strong correlations between parameters, which a diagonal mass matrix cannot remove. 800 leapfrog steps then move the soft

directions by only a fraction of their width per iteration. There is a further trap: the mass matrix is estimated from the spread of the chain during warm-up, so a direction that barely moves gets a small estimated variance and moves even less in the next window.

3.4 The likelihood is too confident

At the end of the run, $U \approx 163{,}900$. The prior term $\frac{1}{2}\sum_i \tau_{g(i)}\omega_i^2$ is about $p/2 \approx 2{,}000$, the rest is the likelihood term $\frac{1}{2}\sum_k r_k^2$: $\chi^2 \approx 1.18$ per training row (with the training variances, which include the one-event term and are therefore, if anything, generous), where a correct noise model with $\gamma \ll N$ well-determined parameters should give $1 - \gamma/N \approx 1$. The training data themselves therefore say what the test spectra said: the residual variance is about 20% larger than the POWHEG noise, on average over m_{hh} (cause C).

4 Step 1: the linearised Laplace approximation

4.1 What it is

Near a mode $\hat{\boldsymbol{\omega}}$, replace the network by its linearisation $f(\boldsymbol{x};\boldsymbol{\omega}) \approx f(\boldsymbol{x};\hat{\boldsymbol{\omega}}) + \boldsymbol{j}(\boldsymbol{x})^{\mathsf{T}}(\boldsymbol{\omega} - \hat{\boldsymbol{\omega}})$, with $\boldsymbol{j} = \partial f/\partial\boldsymbol{\omega}$. The posterior is then Gaussian, with precision (the generalised Gauss–Newton matrix plus the prior)

$$H = \sum_{k=1}^{N} \frac{\boldsymbol{j}_k \boldsymbol{j}_k^{\mathsf{T}}}{\sigma_k^2} + \operatorname{diag}(\tau_{g(i)}), \tag{2}$$

and the predictive standard deviation at any point is

$$s(\boldsymbol{x}) = \sqrt{\boldsymbol{j}(\boldsymbol{x})^{\mathsf{T}} H^{-1} \boldsymbol{j}(\boldsymbol{x})}. \tag{3}$$

This is MacKay's original approach to Bayesian networks [1], revived in recent years as the "linearised Laplace" or "GLM predictive" [5, 6]. It needs no sampling: one pass over the training rows to build H, a Cholesky factorisation, and one pass over the points of interest.

Why it is cheap here

H is $3{,}944 \times 3{,}944$: 15.6 million numbers, 125 MB in double precision. Each row's gradient $\boldsymbol{j}_k$ is obtained with `torch.func` (`vmap` of `grad`) for a chunk of a few thousand rows at a time; the chunk's contribution $J^{\mathsf{T}}WJ$ is one matrix product. The 274,152 rows take of order a minute on the 5090; the Cholesky factorisation a second.

4.2 What it delivers

1. The within-mode width s_k on the test spectra, to be compared with the chain's $0.005\,\sigma_k$ and the ensemble's $0.056\,\sigma_k$.
2. $\gamma = \operatorname{tr}(J^{\mathsf{T}}WJ\,H^{-1})$ exactly, eq. (1).
3. The eigenvalues of H: its condition number tells us how badly a diagonal mass matrix must fail, and how long chains would have to be.
4. The mass matrix for Step 2.

Where to expand

The expansion point should be a mode of the *posterior*, i.e. a minimum of the χ^2 plus the prior. The fine-tune with $\delta = \infty$ (`runs/FT_chi2`, from `code/heft_finetune.ipynb`; it is not in the Mac's `runs` folder, so it is presumably on the 5090) is the χ^2 minimum; a short polish with the prior term added (the fine-tune with a weight penalty $\frac{1}{2}\sum_i \tau_{g(i)}\omega_i^2$, using the precisions of the last Bayesian sample) turns it into the posterior mode. We also expand about each of the 20 fine-tuned ensemble members ($\delta = 2$): this gives a *mixture of Laplace approximations* [7], a cheap preview of Steps 2–3.

Decision rule after Step 1

- Laplace width $\gg 0.005\,\sigma_k$ (we expect a few per cent of σ_k): cause A confirmed; go to Step 2.
- Laplace width $\approx 0.005\,\sigma_k$: the mode really is that narrow; the sampler is not the problem, and the answer must come from B and C (Steps 3–4).

Limits of the approximation

The Gauss–Newton matrix drops the term $\sum_k r_k\,\partial^2 f_k/\partial\omega^2$; with r_k of order the noise and the model near a mode, that term is small, and dropping it keeps H positive. The linearisation is only as good as the network is linear over the width of the mode, which is exactly what Step 2 tests.

5 Step 2: HMC with a dense mass matrix

If the momenta are drawn with covariance $M = H$ (eq. (2)), the Hamiltonian dynamics see a posterior that is, near the mode, a unit sphere: all directions have the same curvature, the step size can be of order 0.1–0.5, and a trajectory of a few tens of steps crosses the mode [3, 4]. Concretely, with the Cholesky factor $H = LL^\mathsf{T}$:

$$\boldsymbol{p} = L\boldsymbol{z},\ \ \boldsymbol{z} \sim \mathcal{N}(0, I); \qquad K(\boldsymbol{p}) = \tfrac{1}{2}\boldsymbol{p}^\mathsf{T} H^{-1}\boldsymbol{p}; \qquad \boldsymbol{\omega} \leftarrow \boldsymbol{\omega} + \varepsilon\, H^{-1}\boldsymbol{p}, \tag{4}$$

where $H^{-1}\boldsymbol{p}$ is two triangular solves with L. Each costs about 1.6×10^7 operations, negligible beside one gradient over 274,152 rows.

- **Start** at the posterior mode of Step 1, not at the L1 solution: the warm-up then adapts the step size at equilibrium instead of during a descent.
- **Warm-up:** step size only (dual averaging); the dense mass matrix is fixed. Optionally recompute H once at the end of warm-up, at the current point and with the current precisions τ_g.
- **Leapfrog steps:** jittered around a trajectory length of order $\pi/2$ in the whitened units, i.e. $L \approx (\pi/2)/\varepsilon$; tens, not hundreds. Each iteration is therefore much cheaper than now, and chains can be much longer for the same GPU time.
- **The Gibbs updates of the** τ_g change the prior part of the posterior's curvature; with the likelihood dominant this matters little, and the sampler stays correct for any fixed M, only its efficiency depends on it.

Success criteria for one chain

No drift of U in the sampling phase; within-chain width at the test points close to the Laplace width; split-$\hat{R} < 1.01$ [11] and an effective sample size of several hundred for predictions at many points (not two), and for U; ideally the same width whether the chain starts at the mode or at a point displaced by a draw from the Laplace approximation.

6 Step 3: several chains

Run 4 to 8 chains, each with its own seed, started at the posterior modes polished from different fine-tuned ensemble members (Step 1). The chains are independent processes (separate `tmux` windows on the 5090, each with its own run folder); memory is not a constraint. For each prediction we then have the within-chain variance W and the between-chain variance B.

- $\hat{R}$ **on predictions** at a few thousand points: if $\hat{R} \gg 1$ the chains are in different modes, which no single chain will leave.
- **Do the modes matter?** Compare the chains' average U and, more completely, their Laplace evidences $\ln Z_m \approx -U(\hat{\omega}_m) + \frac{1}{2} p \ln 2\pi - \frac{1}{2} \ln\det H_m$. Differences of order a few units: the modes carry comparable mass and the posterior is their mixture with roughly equal weights. Differences of tens to hundreds: the posterior is concentrated on the best mode(s).
- **Combining non-mixing chains.** Equal weights are correct only in the first case. Otherwise the chains are weighted by stacking, which chooses the weights that maximise the predictive density on held-out data [9]; the 38,076 unused training rows serve for that.

What this can and cannot establish

If the mixture of chains reproduces the ensemble spread, the Bayesian and ensemble uncertainties agree on the variability of the network. If the posterior is dominated by one mode, the Bayesian answer is legitimately narrower than the ensemble: it then describes the uncertainty of the best solution found, while the ensemble describes the variability of the training procedure. Both are honest; they answer different questions, and the paper must say which question each answers.

7 Step 4: a likelihood that admits the model error

Replace the noise variance by $(1 + a_r)\,\sigma_k^2$, where r labels a few regions of m_{hh} (two, as in the appendix, or eight), and sample $a_r \geq 0$ along with the network. Since the likelihood is Gaussian, the update of $s_r^2 = 1 + a_r$ is a conjugate Gibbs step: with an inverse-Gamma prior $\mathrm{IG}(\alpha_0, \beta_0)$,

$$s_r^2 \sim \mathrm{IG}\Big(\alpha_0 + \tfrac{1}{2} N_r,\ \beta_0 + \tfrac{1}{2} \sum_{k \in r} r_k^2\Big) \text{ truncated to } s_r^2 \geq 1, \tag{5}$$

where N_r is the number of training rows in region r and r_k the pulls. The machinery exists already (`GaussianLikelihood` samples one scale this way).

The Bayesian predictive distribution of the *true* cross section in bin k then has variance $\mathrm{Var}_{\mathrm{post}}[f_k] + a_r \sigma_k^2$, and that of a new POWHEG spectrum adds σ_k^2; this is the Bayesian counterpart of the model-error term of the appendix, learned from the training data. In addition, the posterior of the network itself widens by $\sqrt{1 + a_r}$, since the data are now weighted less.

The independence assumption

The term $a_r \sigma_k^2$ treats the model error as independent from bin to bin and from one point θ to the next. It is not: the network's error is a smooth function, correlated over neighbouring bins and parameter points. Treating it as independent overstates the information in the data, so the sampled a_r will be determined too precisely, and the value learned from the training rows, to which the network has adapted, need not equal the value needed on new data. (On average the two are similar: about 0.18 from the training rows, section 3, and 0.09–0.29 from the held-out rows.) The principled remedy, a Gaussian-process discrepancy term [10], is beyond this project. We will compare the sampled a_r with the held-out values and, if they differ, say so and use the held-out calibration, as in the appendix.

8 Step 5: sensitivity to the prior

With the Laplace machinery, the prior's influence on the width costs one Cholesky factorisation per variant: change $\mathrm{diag}(\tau_g)$ in eq. (2) and recompute eq. (3). Variants: (i) the τ_g fixed at their posterior means; (ii) all τ_g ten times smaller (a vaguer prior) and ten times larger; (iii) a different grouping of the parameters. If the width at the test points changes by less than its statistical uncertainty, cause D is excluded; only if it changes appreciably is a chain with a modified prior worth running.

9 What could go in the paper

Outcome	Statement
Laplace and dense-mass HMC agree; mixture of chains $\approx$ ensemble spread; with a_r the pulls are calibrated	"A Bayesian treatment, with a likelihood that includes a model-error term, gives uncertainties consistent with the ensemble and calibrated on the test spectra." The strongest outcome.
As above, but one mode dominates	"The posterior concentrates on one solution; its width describes that solution, while the ensemble also captures the variability between solutions. Including the model error, both are calibrated."
Sampled a_r smaller than the held-out values	As above, with the held-out calibration of the model error, and a sentence on correlated model errors [10].
None of the above converges	Present the posterior mean only, as a cross-check of the central value, and state that a converged posterior for 3,944 parameters and 2.7×10^5 rows was beyond the computing budget.

Table 2: Possible outcomes and the statement each supports.

In every case the paper should report the diagnostics (effective sample sizes, $\hat{R}$, the absence of drift), since a reader who has seen over-confident Bayesian networks before will look for them first; and in no case should it present the current width of $0.005\,\sigma_k$ as "the Bayesian uncertainty".

10 Code

- **New module** `dihiggs/laplace.py`, heavily commented like the others:

```
H = gauss_newton(model, X, V, prior_precision, chunk_size=4096)  # eq. (4)
chol = torch.linalg.cholesky(H)
s = predictive_sd(model, X_test, chol)                            # eq. (5)
gamma = effective_parameters(model, X, V, chol)                   # eq. (3)
lnZ = log_evidence(U_at_mode, chol)
```

- `dihiggs/hmc.py`**:** an option `mass='dense'` with `mass_matrix=H` (Cholesky factor stored; momenta, kinetic energy and position update as in section 5; warm-up adapts the step size only); checkpoints save the factor. The diagonal mode is unchanged. A model-error option for the `poisson` likelihood: `PoissonLikelihood(variance, regions=..., learn=True)`, with the truncated inverse-Gamma Gibbs step.
- **New notebook** `code/heft_laplace.ipynb`**:** Step 1 and Step 5; writes `runs/LAPLACE/` with H, its eigenvalues, γ, the widths and, for each expansion point, the polished mode.
- `code/heft_train_bnn.ipynb`**:** new switches `START` (a parameters file, e.g. a polished mode), `MASS` (`'diag'` or `'dense'`), `MASS_FILE`, `CHAIN` (an index that sets the seed and the run folder `runs/BNN_dense_chain<k>`), `MODEL_ERROR_REGIONS`.
- **New notebook** `code/heft_bnn_chains.ipynb`**:** combines the chains: $\hat{R}$, W and B, evidences, stacking weights, and the calibration table and plots in the format of the appendix.
- **Toy test** in `code/hmc_toy.ipynb`: the dense mass matrix on a strongly correlated Gaussian, where the exact answer is known, before any real run.

11 Order of work

1. Step 1 and Step 5 (Laplace and prior sensitivity): minutes of GPU time; I can run them on the CPU here if you prefer, from the files in `runs/`. *Gate:* the decision rule of section 4.
2. Dense mass matrix in `hmc.py`, toy test, one chain from the polished mode. *Gate:* the success criteria of section 5.
3. 4–8 chains (Step 3), then the same with the model-error term (Step 4).
4. Write-up as `docs/bnn/bnn_run4.pdf`, and a revised Bayesian paragraph for the appendix.

12 Decisions for you

Decisions before I proceed

1. **Laplace first, as a gate** before any new HMC code. Agreed?
2. **Expansion points:** the posterior mode polished from `runs/FT_chi2`, and modes polished from the 20 fine-tuned members of the bootstrap ensemble (on the Mac now `runs/ENS_bootstrap_huber_d2_1`). Agreed?
3. **Dense mass matrix** from the Laplace H, fixed during warm-up (step size only), rather

than a dense matrix estimated from the chain. Agreed?

4. **Number of chains:** 4 or 8?
5. **Model-error regions:** two (as in the appendix) or eight bands of m_{hh}?
6. **Where Step 1 runs:** on the 5090 (a notebook you run) or here, on the CPU, with the parameter files I already have?
7. **Location of this document:** `docs/bnn/bnn_plan2.pdf`?

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